

calculators 73
 estimating 39
 Pythagorean Theorem 129
 surds 40–41
 square units 28, 132
 squared numbers 241
 powers 36
 quadratic equations 192
 squared variables
 quadratic equations 190
 quadratic expressions 176
 squares
 area of quadrilaterals 132
 calculators 73
 geometry 81
 polygons 134
 quadrilaterals 130, 131
 symmetry 88, 89
 tessellations 99
 squaring
 expanding expressions 174
 Pythagorean Theorem 128
 standard form 42–43
 statistics 200–227
 stem-and-leaf diagrams 221
 straight lines 86–87
 angles 85
 subject of a formula 177
 substitution
 equations 180, 186, 187, 192
 expressions 173
 quadratic equations 192
 simultaneous equations 186, 187
 subtended angles 144, 145
 subtraction 17
 algebra 169
 binary numbers 47
 calculators 72
 expressions 172
 fractions 53
 inequalities 198
 negative numbers 34
 positive numbers 34
 vectors 96
 sums 16
 calculators 72, 73
 multiplication 18, 19
 supplementary angles 85
 surds 40–41
 surface area
 cylinder 175
 solids 152, 156–57, 250
 surveys, data collection 204–205
 switch, mental mathematics 69
 symbols 240
 algebra 168

cube roots 37
 division 22
 expressions 172, 173
 greater than 198
 inequality 198
 less than 198
 numbers 15
 ratio 56, 106
 square roots 37
 triangles 116
 see also signs
 symmetry 88–89
 circles 138

T

table of data 226
 pie charts 210
 tables
 data collection 203, 204, 205, 208
 frequency 206, 207, 216
 proportion 58
 taking away (subtraction) 17
 tally charts 205
 tangent formula 161, 162, 163, 164, 165
 tangents 138, 139, 148–49
 calculators 73
 tax 74
 temperature 35
 conversion graph 185
 conversion tables 243
 measurement 242
 tens
 addition 16
 decimal numbers 44
 multiplication 21
 subtraction 17
 tenths 44
 terms
 expressions 172, 173
 moving 178
 sequences 170
 tessellations 99
 thermometers 35
 thousands
 addition 16
 decimal numbers 44
 three-dimensional bar chart 208
 three-dimensional shapes 152
 symmetry 88, 89
 time measurement 28, 30–32, 242
 speed 29
 times tables 67, 241
 tonnes 28

top-heavy fractions 49, 50, 54
 transformations
 enlargements 104
 reflections 102
 rotation 100
 translation 98
 translation 98–99
 transversals 86, 87
 trapezium (trapezoid) 130, 131, 134, 249
 tree diagrams 238–39
 triangles 116–17, 246
 area of 122–24, 249
 calculators 73
 congruent 112, 133
 constructing 118–19
 equilateral 113
 formulas 29, 177
 geometry 81
 parallelograms 133
 Pascal's triangle 247
 polygons 134, 135
 Pythagorean Theorem 128, 129, 249
 rhombus 133
 right-angled 73, 83, 95, 117, 128, 129, 148, 161, 163, 164, 165
 set squares 83
 similar 125–27
 symmetry 88, 89
 tangents 148
 trigonometry formulas 161, 163, 164, 165
 vectors 95, 97
 triangular numbers 15
 trigonometry 158–65
 calculators 73
 formulas 161–65, 248
 turns, angles 84
 24-hour clock 32
 two-dimensional shapes,
 symmetry 88, 89
 two-way table 205

U

units of measurement 28–29, 242
 cubed 154
 ratios 57
 squared 132
 time 30
 units (numbers)
 addition 16
 decimal numbers 44
 multiplication 21
 subtraction 17

unsolvable simultaneous,
 equations 189

V

variables
 equations 180
 simultaneous equations 186, 187
 vectors 94–97
 translation 98, 99
 vertex (vertices) 116
 angles 85
 bisecting an angle 112
 cyclic quadrilaterals 147
 polygons 134
 quadrilaterals 130, 147
 solids 153
 vertical coordinates 90, 91
 vertical (perpendicular) height
 area of quadrilaterals 132, 133
 area of triangles 122, 123
 volumes 155
 vertically-opposite angles 87
 volume 152, 154–55
 conversion tables 243
 density 29
 measurement 28, 242, 250

W

wages 74
 watches 32
 weeks 30
 weight measurement 28
 weighted mean 217

X

x axis,
 bar graphs 206, 207
 graphs 92

Y

y axis
 bar graphs 206, 207
 graphs 92
 yards 28
 years 28, 30

Z

zero 14, 34
 zero correlations 227
 zero power 38